IMRAN PASHA

Backend Engineer | Python, FastAPI, PostgreSQL, Redis, Node.js

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Nationality: Indian | Employment visa via employer sponsorship (standard UAE process) | Languages: English (professional working)

PROFESSIONAL SUMMARY

Backend engineer with nearly two years building production REST APIs and distributed systems in Python (FastAPI) and Node.js (Express). Founding backend engineer on Kaizer X, an AI video-production SaaS live at kaizerx.com, and sole backend engineer on a payments-heavy field-service platform. Designs for failure rather than the happy path: durable PostgreSQL and Redis job queues, idempotency keys, plan-aware rate limiting and process isolation that keeps live broadcasts running through a deploy. Certified in cyber security and ethical hacking through IIT Roorkee's iHUB DivyaSampark, and applies it directly to API design.

TECHNICAL SKILLS

Languages: Python 3.12, JavaScript (Node.js 20), TypeScript, SQL

Backend: FastAPI, Express.js, Django, SQLAlchemy 2.0 (async), Pydantic v2, Alembic, TypeORM, Sequelize, Electron

Data & Caching: PostgreSQL 17 (composite indexing, query-plan tuning, advisory locks, FOR UPDATE SKIP LOCKED), Redis 7 (Streams, consumer groups, Lua scripting), Supabase, SQLite

Architecture: REST API design, OpenAPI/Swagger, microservices, event-driven workflows, message queues, WebSockets, Server-Sent Events, asyncio, background workers, idempotency keys, horizontal scaling

Cloud, DevOps & Observability: AWS (EC2, S3, Lambda), Cloudflare R2, Docker, NGINX, Railway, Render, GitHub Actions CI/CD, GitFlow, Linux, Prometheus, OpenTelemetry, pytest, Jest, load testing

Security: OAuth 2.0, JWT, RBAC, OWASP Top 10 and API Security Top 10, threat modeling (STRIDE), secure code review, rate limiting, HMAC webhook verification, encryption at rest, SSRF/IDOR/BOLA mitigation, VAPT, Burp Suite

AI & Integrations: Google Gemini, Anthropic Claude, OpenAI, Deepgram ASR, FFmpeg/NVENC, YouTube Data API, Meta Graph API, Razorpay, Firebase Cloud Messaging

PROFESSIONAL EXPERIENCE

Backend Engineer | Sharkify Private Limited — Dec 2024 – Present
Hyderabad, India

Kaizer X — AI video production and multi-channel publishing SaaS — Apr 2026 – Present
Python 3.12, FastAPI, SQLAlchemy 2.0, PostgreSQL 17, Redis 7, Docker, Cloudflare R2, FFmpeg/NVENC, Gemini, Claude

- Founded the backend, rewriting a Django prototype as **FastAPI and SQLAlchemy** and owning it solo for five months; it now serves 765 REST and WebSocket routes over 150 PostgreSQL tables.
- Designed a **durable publish queue** on PostgreSQL FOR UPDATE SKIP LOCKED with lease fencing, heartbeat renewal, priority aging and jittered backoff, so workers on separate hosts claim each job exactly once.
- Built a **Redis Streams** consumer-group queue with priority lanes, XCLAIM crash recovery and a dead-letter stream, and killed duplicate publishes with SHA-256 idempotency keys arbitrated by a UNIQUE constraint.
- Orchestrated the AI media pipeline: dual-engine speech-to-text with language failover, a provider-agnostic LLM layer over Gemini, Claude and OpenAI with circuit breakers, and FFmpeg/NVENC rendering.
- Re-architected live streaming into separate API, control, relay and encode processes on a dedicated Redis, so restarting the web tier no longer interrupts a broadcast already on air.
- Instrumented 14 **Prometheus** metrics and OpenTelemetry tracing, then wrote a load harness whose findings cut p95 latency from 12 s to **141 ms** by right-sizing the synchronous-route threadpool.

Kaizer X Desktop — Windows edition (sole author, 163 commits) — Aug 2026 – Oct 2026
Node.js, Electron 33, PyInstaller, Cloudflare R2

- Shipped an **Electron** shell that spawns and supervises a frozen FastAPI backend on localhost, with free-port probing, health-polled startup, orphan reaping and process-tree shutdown.
- Designed an offline-tolerant licensing state machine (12-hour heartbeat, 72-hour grace, lockdown on 403) and streamed multi-gigabyte uploads straight to Cloudflare R2 without buffering them in memory.

Voter Connect — campaign management platform (live on Google Play) Dec 2025 – Jun 2026

FastAPI, PostgreSQL, Redis, async I/O, SQLAlchemy, JWT, Docker, GitHub Actions

- Built and tuned a backend over a **9.9M-row PostgreSQL** dataset, holding sub-second API latency through composite indexing, query-plan analysis and Redis caching of hot read paths.
- Implemented async I/O, JWT authentication and rate limiting, designed geolocation-verified endpoints separating administrator from volunteer authorization, and delivered CI/CD on GitHub Actions with GitFlow.

Editor Dashboard and KaizerFit — FastAPI services Jun 2025 – Oct 2025

FastAPI, async SQLAlchemy, asyncpg, PostgreSQL, Alembic, Server-Sent Events, Telegram Bot API, Docker, AWS EC2

- Built a newsroom submission CMS as principal author (98% of the code by git blame): 54 endpoints, revocable JWTs, and real-time dashboards without polling via PL/pgSQL triggers, pg_notify and **Server-Sent Events**.
- Led a fitness backend (47 endpoints, 21 tables) whose **offline-first sync engine** ingests five record types in one atomic bulk transaction, with ID mapping and hash-based deduplication so client retries never double-count.

Ozone Wash — field-service booking, payments and compliance platform Mar 2026 – Oct 2026

Node.js 20, Express 4, PostgreSQL (Supabase), Cloudflare R2, Razorpay, PayU, Easebuzz, node-cron, Render

- Architected and built the **Node.js/Express** backend as sole backend engineer for six months: 23 feature modules, roughly 200 REST endpoints and 37 transactional SQL migrations defining 68 tables.
- Integrated three payment gateways behind one facade with raw-body **HMAC-SHA256 webhook verification**, replay deduplication and idempotent settlement, so a redelivered webhook never double-confirms a booking.
- Designed a reserve-pay-confirm booking flow whose payment hold is swept by cron and reinstated under a PostgreSQL advisory lock with a capacity re-check, eliminating double-booked crews.
- Built an admin security plane: bcrypt credentials, eight-hour JWTs bound to hashed server-side sessions for instant revocation, five-tier RBAC, Helmet, a CORS allowlist and three rate-limit tiers.

Bose American Academy — learning platform API (2nd of 9 contributors) Dec 2024 – Apr 2025

Node.js, TypeScript, Express, InversifyJS, Sequelize to TypeORM, MSSQL to PostgreSQL (AWS RDS), AWS S3, PayU

- Built the user-identity domain of a **TypeScript/Express** API using dependency injection: registration with email one-time passcodes, multi-role login, three-step password reset and Google OAuth 2.0 code exchange.
- Implemented PayU post-payment fulfilment — gateway re-verification, voucher consumption, cart clearing and course enrolment — then extended checkout to multi-course orders.

SECURITY ENGINEERING APPLIED TO BACKEND

Cyber security training applied directly to the systems above, not kept in a separate track.

- Closed a two-round adversarial security review with a regression test behind every fix: cross-tenant authorization (**BOLA/IDOR**) on live-broadcast controls and WebSockets, OAuth-callback cross-site scripting, and password-reset links trusting an attacker-controlled Origin header.
- Stopped a third-party API key leaking through exception text on an unauthenticated route, adding a central secret-redaction module applied across logs and error paths.
- Wrote an **SSRF guard** that resolves DNS before refusing loopback, private, link-local and cloud-metadata addresses, and replaced client-declared MIME types with byte-level magic-number verification.
- Applies **OWASP API Security Top 10**, STRIDE threat modeling and secure code review from an IIT Roorkee programme and 49 hands-on lab rooms, treating authorization and rate limits as design inputs rather than afterthoughts.

EDUCATION

B.Tech, Computer Science | Sahyadri College of Engineering (VTU), Mangalore 2020 – 2024

CERTIFICATIONS

Executive Post Graduate Certification in Cyber Security and Ethical Hacking | IIT Roorkee May 2026

iHUB DivyaSampark, IIT Roorkee, in association with Intellipaat. Credential ID IPTIH26050044. 49 hands-on lab rooms: web application testing, Burp Suite, Nmap, Metasploit, privilege escalation, SQL injection, cross-site scripting and IDOR.

SELECTED ACHIEVEMENTS

- Diagnosed a production queue runaway that grew one Redis stream from 8 to **619,000 entries**, traced it to a time-of-check-to-time-of-use race and fixed it with an atomic conditional claim.
- Root-caused a two-hour streaming degradation that every internal signal called healthy to a **0.72 Mbps host uplink**, then added byte-delivery accounting and capacity-based admission control.
- Delivered two Google Play applications and five production backend platforms across Python and Node.js in under two years.